

Current-Source Inverter Integrated Motor Drives Using Dual-Gate Four-Quadrant Wide-Bandgap Power Switches

Renato Amorim Torres , *Member, IEEE*

almost twice the power density (5.9 kW/kg) of the best machine designed for voltage source-based drives (97.3%, 3.1 kW/kg).

Therefore, the long-neglected CSI topology offers advantages by reducing EMI noise, avoiding motor terminal overvoltages, replacing the VSI's temperature-limited dc-link capacitor with a more robust dc-link inductor, providing boost capability, and improving voltage and current total harmonic distortion at the machine terminals [2], [3].

One of the major drawbacks of CSIs is the requirement to connect a power diode in series with the switching device (switch + diode) to provide RVB capability, which significantly increases the inverter's conduction losses. Devices with natural RVB capability can substantially improve the CSI's efficiency by eliminating the series diodes. When WBG devices are designed as lateral devices (typical for gallium nitride (GaN) devices, but also conceivable in SiC [7]), their channels are symmetrical, making them candidates for monolithic devices with four-quadrant (FQ) capabilities that include RVB properties.

In [8], an appealing monolithic bidirectional switch is designed and fabricated in a common-drain configuration, which substantially reduces its on-resistance and size. In [9], a different monolithic FQ switch topology that also adopts a common-drain structure is proposed. The conduction loss in a CSI power stage that uses monolithic FQ switches in common-drain topologies that share their drift regions can achieve approximately half the conduction loss of a CSI using hybrid switches [10] and even a lower fraction when compared to the baseline CSI using the switch + diode configuration. The use of FQ switches in CSIs also makes it possible to reverse the dc-link current polarity. This feature is particularly advantageous in systems with distributed dc voltage architectures, where the dc-link current polarity can be used to reverse the power flow direction, eliminating the need for full-bridge front-ends [10]. However, FQ switches present challenges during switching events due to problems caused by either dead-time or conduction overlap when commutating the load current [11]. Therefore, the introduction of FQ switches into CSIs requires further investigation since it cannot serve as a simple drop-in replacement for the baseline switch + diode configuration.

FQ switches have received more attention during recent years due to their appeal for use in several power converter topologies, including the matrix converter, Vienna rectifier, T-type multi-level converters, and cyclo-converters. For example, Panasonic has already released research samples of GaN-based monolithic FQ switches that can block voltage >1300 V in both polarities while providing ON-state resistance values similar to those of conventional switches [12].

Recent studies have evaluated many implementation issues associated with WBG switches, including gate drive circuits, dead-time, and parasitic effects [13]–[15]. However, only a few papers have evaluated FQ switches implemented with WBG devices and their use in CSIs. In [3], potential benefits of using FQ switches in CSIs, such as improved efficiency, are identified. Although [3] includes the characterization of a hybrid FQ switch using discrete devices, the article does not focus attention on the issues associated with using FQ bidirectional switches, and no full inverter operation results are provided. In [10], monolithic

FQ switches are used to improve a buck-boost current-source inverter's efficiency, power density and allow bidirectional power flow; however, the article is mainly focused on analyzing a new modulation strategy that reduces the conduction losses and switching losses by shaping the dc-link current. In [16], a new fault-tolerant CSI topology using FQ switches is proposed. The topology takes advantage of monolithic FQ switches to achieve redundancy, fault detection, and loss reduction via current sharing between parallel devices. In [17], FQ switches are employed in the CSR stage of an indirect matrix converter; however, the article is mainly focused on control strategies for operation under unbalanced grid voltage conditions. Neither [10], [16], or [17] analyze the commutation of FQ switches in CSIs and strategies to minimize their losses.

The major objective of this article is to demonstrate that the CSI topology in combination with FQ switches can provide high efficiency and high-power density, making this topology a serious candidate for adoption in IMDs. This article presents, analyzes, and implements commutation methods tailored for applying FQ switches in CSIs that achieve safe and high-efficiency operation. The analytical results are experimentally verified using a 3 kW CSI-IMD demonstrator unit exciting a permanent magnet machine.

The article is structured as follows. In Section II, an explanation of the advantages of using FQ switches in CSIs is provided and different FQ switches realizations are discussed. Section III provides a detailed explanation of the four-step commutation method for CSIs using FQ switches. Key implementation issues are also discussed in detail. Section IV introduces a three-step improved commutation method that reduces the switching loss by taking advantage of the knowledge of the output capacitors' voltage polarity. Section V presents extensive experimental results, including results obtained from a bench-top CSI driving an *RL*-load, and results from a CSI-IMD exciting a surface permanent magnet (SPM) motor. Finally, Section VI uses the experimental results to create efficiency maps for alternative topologies and to estimate the efficiency of future CSI-IMDs that employ monolithic bidirectional switches.

II. HYBRID SWITCH REALIZATIONS

As discussed in the preceding section, the use of monolithic FQ switches is notably beneficial for CSIs since it can significantly reduce conduction losses. However, such monolithic devices are still commercially immature. In their absence, hybrid switches consisting of two discrete devices connected in anti-series can be used to mimic the behavior of future monolithic FQ devices. Four possibilities for the hybrid switch realization using anti-series WBG switches are presented in Fig. 1. Fig. 1(a) and (b) use WBG devices such as lateral GaN high-electron-mobility transistors (HEMTs) that do not incorporate an intrinsic body diode. Two of these devices can be connected in either a common-drain [Fig. 1(a)] or common-source configuration [Fig. 1(b)], both with dual gate drives. Alternatively, WBG devices such as vertical SiC MOSFETs that include body diodes can be used in either the common-drain or common-source configurations, as shown in Fig. 1(c) and (d), respectively. Note

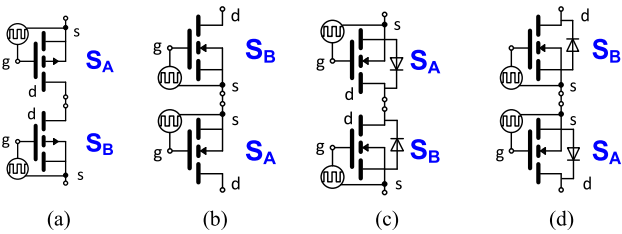
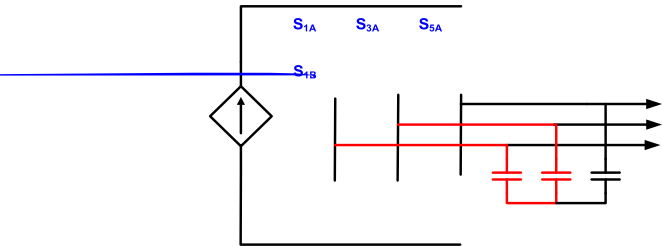
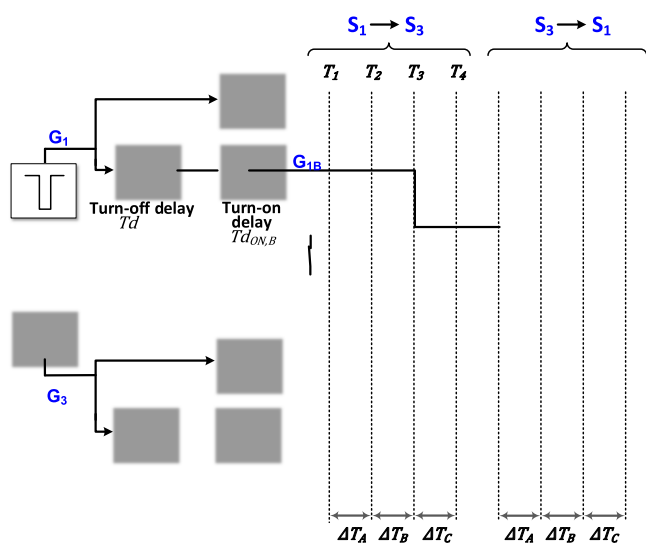


Fig. 1. FQ hybrid switches using two anti-series devices without body diodes. (a) Common-drain configuration. (b) Common-source configuration. FQ hybrid switches using anti-series devices with body diodes. (c) Common-drain configuration. (d) Common-source configuration. [18].





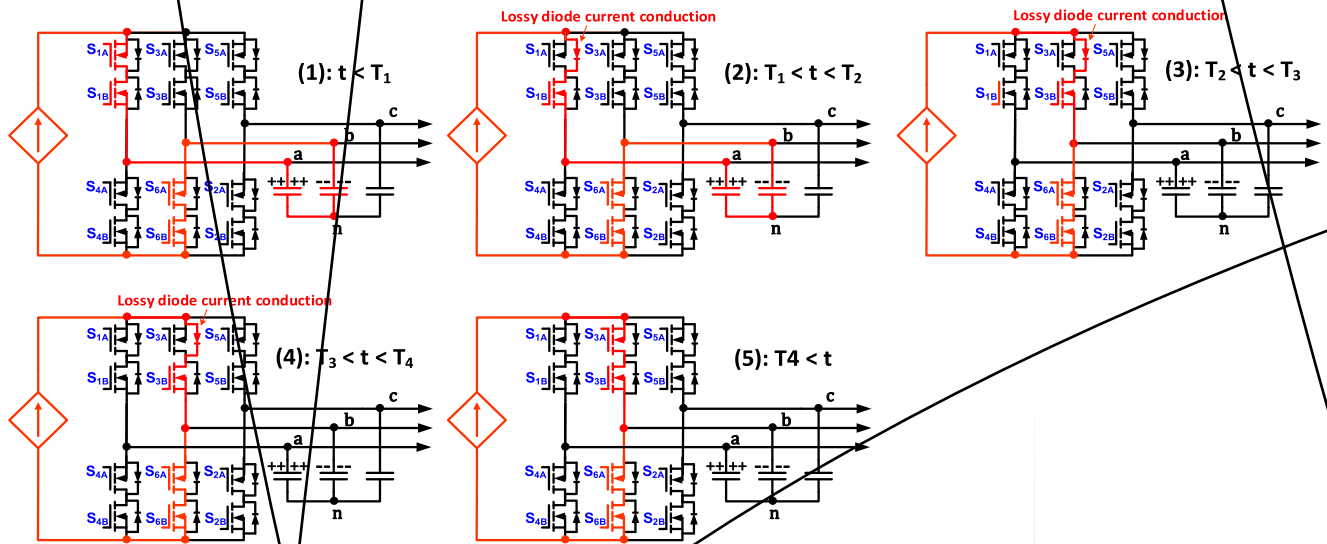


Fig. 5. Step-by-step commutation method for CSIs implemented with FQ switches using four-step algorithm, showing commutation from S_1 to S_3 conduction.

$T_{d\ ON,A}$, the A switch will not even be turned ON, which again does not represent a risk, but only contributes to a slight increase in the conduction losses since no synchronized conduction is performed. In order to minimize

Turn-on
delay
 $T_{d\ ON,A}$

$S_1 \rightarrow S_3$ $S_3 \rightarrow S_1$

tage of guaranteeing the safe commutation of the FQ switches

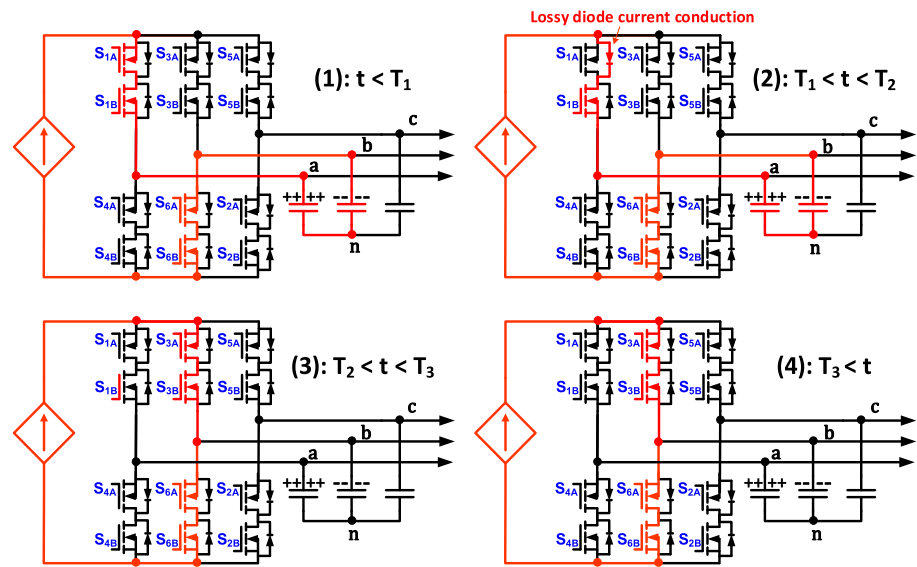


Fig. 7. Step-by-step commutation method for CSIs implemented with FQ switches using three-step algorithm, showing commutation from S_1 to S_3 conduction for $V_{ab} > 0$.

TABLE I
CORRESPONDENCE OF GATE SIGNALS AND SWITCHES

Gate Signal	Discrete Switch	Gate Signal	Discrete Switch
$G_{a,top,swA}$	G_{1A}	$G_{a,top,swB}$	G_{1B}
$G_{b,top,swA}$	G_{3A}	$G_{b,top,swB}$	G_{3B}

TABLE II
ACTIVE AND ZERO STATES FOR DIFFERENT SVPWM SECTORS

Sector	State A	State B	Zero State Option 1	Zero State Option 2
I	S_1, S_2	S_6, S_1	S_3, S_6	S_5, S_2
II	S_2, S_3	S_1, S_2	S_1, S_4	S_3, S_6
III	S_3, S_4	S_2, S_3	S_5, S_2	S_1, S_4
IV	S_4, S_5	S_3, S_4	S_3, S_6	S_5, S_2
V	S_5, S_6	S_4, S_5	S_1, S_4	S_3, S_6
VI	S_6, S_1	S_5, S_6	S_5, S_2	S_1, S_4

Therefore, using this hybrid method, the conditional inequalities that appear in (9) are modified to be:

Case $V_{xy} > e_{\text{margin}} \rightarrow$ Three - step operation

Case $V_{xy} < -e_{\text{margin}} \rightarrow$ Three - step operation. (10)

Equation (11) defines when four-step commutation is applied:

Case $(-e_{\text{margin}} < V_{xy} < e_{\text{margin}}) \rightarrow$ Four - step op. (11)

A. Operation of the Three-Step Commutation When Using Space Vector Pulse Width Modulation (SVPWM)

When using the SVPWM, only four switches are used per sector. For example, during sector II, only switches 1, 2, 3, and 6 operate if the zero-state is realized by turning ON switches 3 and 6. Therefore, during this sector, the commutation among the top switches is always independent of switch 5, which is continuously OFF. Similarly, the commutation among the bottom switches is also independent of switch 4. Consider Fig. 7, which shows a commutation event that occurs within Sector II of the SVPWM. Given the fact that switch 5 never operates within this sector and given the voltage polarization of the output capacitors $V_{ab} > 0$, the discrete switch S_{3A} can be left in the ON-state during the entire sector II without posing the risk of shorting the output capacitors. In summary, instead of turning ON and OFF switch S_{3A} in synchronism with the switch S_{3B} as previously shown in Fig. 6, the switch S_{3A} can be kept on during the entire sector II as long as $V_{ab} > 0$. On the other hand, in case $V_{ab} < 0$, the switch S_{1A} would be the one kept continuously on during sector II.

Keeping the A switch continuously ON within the sector interval avoids desynchronization between the A and B switches due to propagation delays. Such desynchronization can lead to the lossy switch-diode mode operation. For example, suppose that the gate signal G_{3A} shown in Fig. 6 is slightly delayed with respect to the gate signal G_{3B} . The delay would lead the hybrid switch to operate for a short interval in the lossy switch-diode mode before the A-switch is turned ON. This problem is prevented if the gate signal G_{3A} can be safely kept continuously ON within the entire sector (given $V_{ab} > 0$).

Table II shows the active states and the zero state options that ensure that, for every sector of the SVPWM, there will always be two switches constantly OFF (one among the top switches and another among the bottom ones). Therefore, as previously explained, the A switch of the top hybrid switch connected to the output phase of lower voltage and the A switch

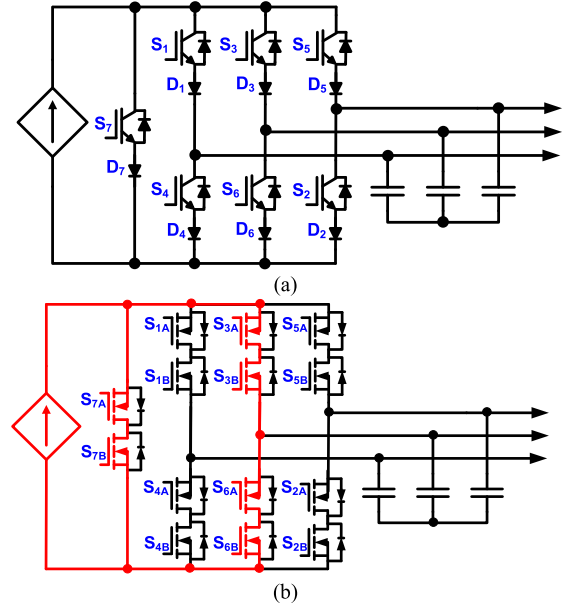


Fig. 9. (a) Three-phase H7-CSI topology using IGBT + diode RVB switches. (b) Three-phase H7-CSI topology implemented with common-drain FQ switches [Fig. 1(c)].

of the bottom hybrid switch connected to the output phase of higher voltage can be kept continuously ON during the entire SVPWM sector interval. Such operation can reduce the CSI losses associated with the switch-diode conduction mode due to desynchronization of the gate signals A and B.

B. Commutation of H7 CSI Implemented With Four-Quadrant Switches

An existing alternative CSI topology that improves the efficiency of the baseline H6-CSI topology is the H7-CSI topology shown in Fig. 9. The additional switch S_7 is used to implement the PWM zero states instead of the S_1 – S_6 switches, allowing these switches to change states under zero-current soft-switching conditions. This section discusses extension of the four-step and three-step commutation methods to this H7-CSI topology when using FQ switches.

The H7-CSI topology using the (switch + diode) configuration for S_1 – S_7 is shown in Fig. 9(a). This topology reduces the conduction losses during zero-states by activating only the S_7 switch instead of both switches in any of the three phase-legs. Fig. 9(b) shows the H7-CSI topology when bidirectional switches are used for the S_1 – S_6 switches. In this topology, the seventh switch must also be implemented as an FQ switch. If a single switch were used without a series diode, the constraint (b) presented in Section III would not be met during active states. More specifically, the output capacitors would be shorted during active states due to conduction of the S_7 body diode.

The S_7 switch can be incorporated into both the four-step and three-step commutation methods, and its operation can be summarized by a fundamental rule for safely operating the seventh switch: if both FQ switches in any inverter phase-leg are completely turned-ON (i.e., a zero state), the S_7 FQ switch

can be active. Applying this rule to the four-step commutation method example in Figs. 3 and 5 means that the S_7 switch will be turned ON at instant T_4 when all four MOSFETs in the S_3/S_6 phase-leg are in their ON-states. Applying this rule to the three-step commutation method example in Figs. 6 and 7 means that the S_7 switch will be turned ON at instant T_2 . As before with the H6-CSI topology, the switching events are symmetrically reversed in time sequence during the turn-OFF process.

Both the S_7 switch and a complete phase-leg of the inverter can be ON at the same time during zero states, as shown in Fig. 9. This reduces the conduction losses by providing two parallel paths for the dc-link current. Assuming that all of the FQ switches have the same channel resistance R , the conduction loss during the zero state can be calculated as

$$P_{\text{cond}} = I_{\text{DC}}^2 (R \parallel 2R) = \frac{1}{3} (2RI_{\text{DC}}^2). \quad (12)$$

That is, the conduction loss during the zero state in the H7 topology is one-third of the H6 zero-state conduction loss. If SVPWM modulation is used, the zero state normalized operating time for one SVPWM sector can be calculated as shown in (13) and (14), according to [21]. Therefore, employing the H7 configuration to reduce the zero state conduction loss up to 66.7% has its largest impact when the modulation index $m_a = 0$, and the impact of this reduction gradually decreases as the modulation index approaches unity, $m_a = 1$.

$$\frac{T_0}{T_s} = 1 - m_a \left(\sin \left(\frac{\pi}{6} - \theta \right) + \sin \left(\frac{\pi}{6} + \theta \right) \right) \quad (13)$$

$$\text{avg} \left(\frac{T_0}{T_s} \right) = 1 - 0.95493 m_a. \quad (14)$$

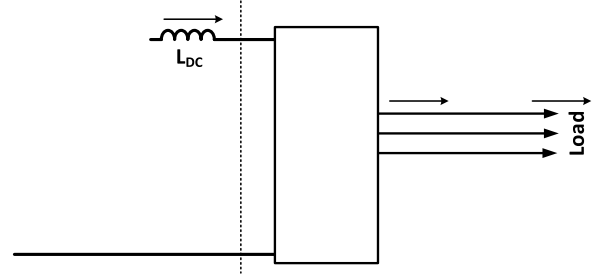
More information about the H7-CSI topology including an alternative modulation strategy that allows the use of single-gate bidirectional switches can be found in [22].

V. EXPERIMENTAL RESULTS

A. Instrumentation and Efficiency Measurements

Fig. 10(b) illustrates the voltage and current waveforms of a typical CSI when exciting an RL load or a motor. The efficiency is measured from points A to B in Fig. 10(a). This setup measures an unfiltered voltage waveform at the dc-link but has the advantage of excluding the dc-link inductor losses from the measurements.

It has been experimentally verified that for both measurement techniques, less than 0.01% of the input and output power content is associated with frequency components higher than 1 kHz. This conclusion is based on experimental data recorded using a *MDA810A* oscilloscope at a sampling rate of 500 MS/s when using current and voltage probes with bandwidths of 100 MHz (CP031) and 120 MHz (HDV3106), respectively. Since nearly all the power content lies at low frequencies below 1 kHz, power measurement instruments that have high accuracy in this low frequency range are preferred. The *Yokogawa 1806T* power analyzer was selected for the CSI efficiency measurement due to the high accuracy of its voltage and current measurements. According to the equipment manual, the power measurement



Test #1: Total

current is not zero but approx. 3 to 3.5 W. This suggests that there is another source of loss that generates considerable loss even at low currents. This source of loss should be independent of current and not associated with switching events since, at this extrapolated point, the current and the switching frequency losses both tend to zero. As will be discussed later, the presence of this term is most likely associated with the output capacitor dielectric losses.

Fig. 11(c) shows that in the FET + FET configurations, the conduction loss has a quadratic behavior with respect to the current since the conduction loss is due to the hybrid switch's channel resistances. In the CSI, at any instant, the dc-link current is being conducted by two hybrid switches consisting of four channel resistances. Based on the 0.36Ω resistance coefficient obtained from the curve fitting [see in Fig. 11(c)], the channel resistance of each device can be estimated to be approx. $90 m\Omega$. This resistance value also accounts for other contributors to the conduction losses including the PCB traces, wires, and connectors. In the FET + diode configuration, the conduction loss function has a first-order term associated with the diode's voltage drop and a quadratic term associated with the resistive components contributed by the diodes and switches.

Fig. 11(d) shows that for all configurations, the switching loss sensitivity to the current level is low. Nevertheless, the FET + FET configuration still has a higher dependence on the current level since, for this configuration, every commutation has intervals of switch-diode conduction mode where the losses are dependent on the current level. This dependence is attenuated when using the three-step commutation method, as expected.

Test #3. Loss sensitivity to output voltage: This test bears some similarity to Test #2, but the switching and conduction losses are evaluated as the voltage level is changed. Due to the limitations of the connected load, this test was realized at a fixed current of 8.5 A, slightly above the current level of 8 A used for baseline operation. The results of Test #3 are displayed in Fig. 11(e)–(g). Contrary to expectations, the conduction losses in Fig. 11(f) increase with voltage. Since the conduction loss should not depend on the voltage, this voltage dependence suggests that there is another loss component beyond the switching and conduction losses that has not been incorporated into the model. Based on Fig. 11(f), this loss component increases with voltage, but it is decoupled from switching events, since Fig. 11(f) shows curves that were extrapolated via linear regression where the switching frequency tends to zero. This stray loss will be examined in the next test. Finally, Fig. 11(g) shows that the switching loss increases with voltage in an approximately quadratic manner.

Test #4. Stray losses: As mentioned in the discussion of results for Tests #2 and #3, the inverter appears to have a loss component beyond the switching and conduction losses that is referred to as stray loss. The stray loss for each configuration can be calculated by subtracting the offset of the conduction loss, as shown by the arrows labeled as “stray loss” in Fig. 11(f). The stray losses calculated for each configuration from the Test #3 data is presented separately in Fig. 11(h). Since the inductor is excluded from the loss measurement, this stray loss is most likely associated with the output capacitors. In fact, the capacitor

dielectric losses can be modeled as (15) [25], where K_{diel} is the dielectric loss constant, V_{out} is the CSI output voltage applied across the capacitor and f is the frequency

$$P_{\text{diel}} = K_{\text{diel}} V_{\text{out}}^2 f. \quad (15)$$

To strengthen the argument that the stray losses are associated with the capacitor dielectric loss, another test was conducted operating the CSI FET + diode topology with the same parameters of the baseline operation but at a fundamental frequency of 25 Hz, which corresponds to 1/4 of the original fundamental frequency. Since the voltage, current, and switching frequency are the same, the difference in the losses would correspond to the difference in the capacitor dielectric loss. Noteworthy, at such low fundamental frequencies, the load reactance is negligible and the change in the fundamental frequency has minimal impact on the load impedance and power factor. Based on (15), the dielectric loss at 25 Hz should correspond to 25% of the dielectric loss at 100 Hz. Therefore, since the estimated dielectric loss at 100 Hz is 3.4 W [see Fig. 11(h) @198.4 V], the stray loss should reduce to 0.85 W, which is close to the 1.1 W measured experimentally. Unfortunately, the majority of ceramic capacitor manufacturers do not provide data for evaluating losses associated with low-frequency, large-signal waveforms [25], [26].

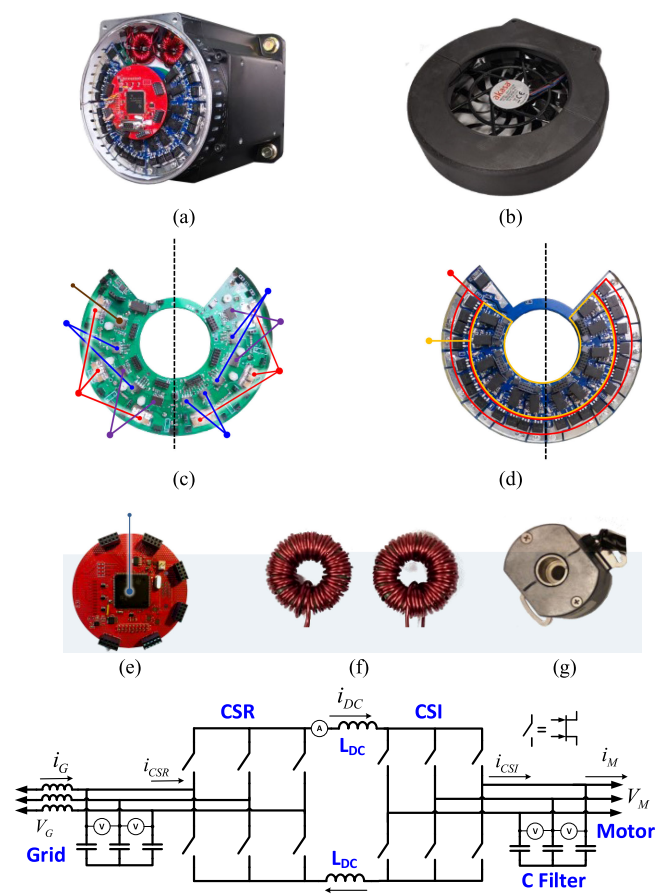
The observed stray loss in the inverter and its attribution to the capacitor dielectric loss deserves further investigation, but this topic falls outside the scope of this article. The authors in [25] and [26] provide techniques for measuring large-signal losses in ceramic capacitors. Finally, some of the observed stray losses are likely a consequence of minor unmodeled loss sources. For example, parasitic resistances, inductances, and capacitances can create resonant circuits that are excited nonuniformly at different switching frequencies, generating low-amplitude losses that were not modeled during this investigation.

Test #5. Loss sensitivity to dead-time: This test is meaningful only for the FET + FET topology. The dead-time interval that defines the duration of the switch-diode mode is varied to evaluate the contribution of the switch-diode conduction losses for both the four-step and three-step commutation methods. The results of this test are presented in Fig. 11(i). As previously explained in Section IV, the three-step commutation method enables a reduction of the switch-diode mode loss by 66.7% compared to the four-step method. Therefore, by varying the dead-time interval, it is expected that the loss variation with three-step commutation will be approx. three times lower than the loss variation with the four-step method. However, even though the three-step commutation reduces the time that the dc-link current is being conducted through the lossy switch body diodes to one-third of the time with four-step commutation, this current is redirected to the switch A channel, which also has some ohmic loss.

The expected loss variation with respect to dead-time for the four-step and for the three-step commutation can be expressed as shown in (16) and (17), respectively. In the four-step commutation, the loss variation is completely associated with the interval when the conduction loss is dissipated in the body diode. On the other hand, with three-step commutation, the loss variation

TABLE V
LOSS COEFFICIENTS FOR CSI LOSS ESTIMATION

Coefficient	FET+Diode	FET+FET	FET+FET	Unit
V_{drop}	1.197	0	0	V
$R_{channel}$	0.3	0.36	0.36	Ω
K_{diel}	0.879	0.358	0.266	$\mu\text{Ws}/\text{V}^2$
$K_{sw,l}$	0.267	0.598	0.379	Ω
$K_{sw,v}$	127	146	179	$\mu\Omega^{-1}$
$L_{sw@2kW}$	5.79	9.50	8.46	W
$V_{out@2kW}$	198.35	198.35	198.35	V
$I_{out@2kW}$	8.40	8.40	8.40	A
$f_{sw_{ba}}$				



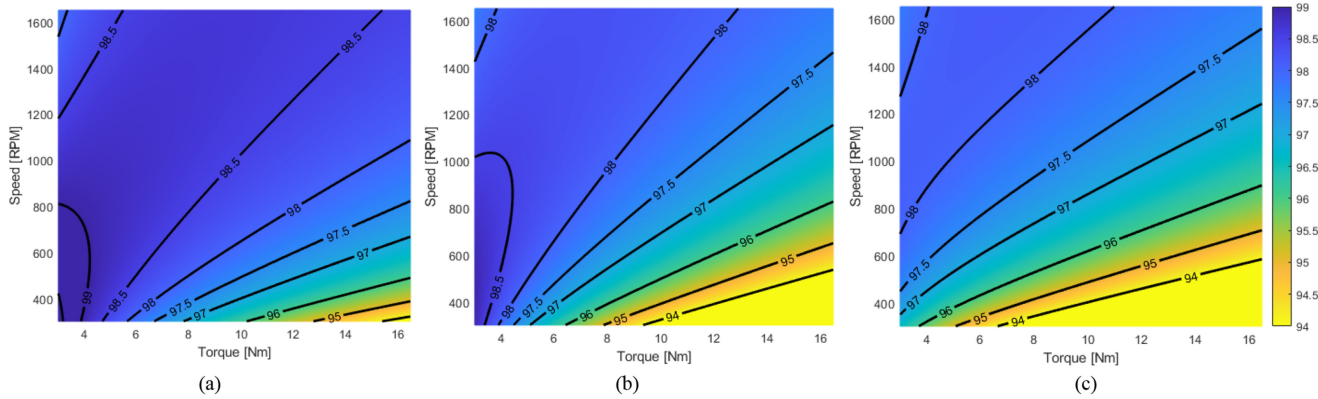


Fig. 14. Predicted CSI efficiency maps for 3 kW unit based on empirical formula (19). (a) Efficiency for CSI-IMD implemented with monolithic bidirectional switches with ON-state-resistance equal to half the ON-state resistance of hybrid bidirectional switches. (b) Efficiency of CSI-IMD using the hybrid FET + FET switch configuration using three-step operation. (c) Efficiency of CSI-IMD using conventional FET + diode switch configuration.

monolithic FQ switches is consistent with results reported in the literature [10].

In contrast to the preceding section, these efficiency maps include losses associated with the *Edge* powder core dc-link inductor. The inductor losses were estimated based on the methods provided in [29] and used the inductor core material curves provided by the manufacturer [30]. High-frequency ac loss components (100 kHz) including skin effect and hysteresis losses are also included in the loss estimates. The torque and speed values are based on the same demonstrator SPM motor described above.

For comparison, Fig. 14(b) and (c) show the efficiency map for the same CSI-IMD using the hybrid FET + FET switch configuration with three-step commutation and for a CSI using conventional FET + diode switches, respectively.

These plots show similar trends for the efficiency that drops considerably for low speeds and high torques. At low speeds, the output power is low at high torque values due to the high conduction losses produced by the high currents needed to produce the torque. Nevertheless, these plots show that adoption of monolithic FQ switches can substantially boost the CSI efficiency over a wide range of torque-speed operating points that is achieved by reducing the conduction loss, which is the main loss component in the CSI. By comparing Fig. 14(a) against Fig. 14(b) and (c), it can be seen that the efficiency improvement obtained when using monolithic FQ switches is more significant at low-speed and high torque operating conditions.

As noted in Section V, the FET + FET configuration has higher switching losses than the FET + diode configuration. Therefore, if the switching frequency is high enough, the traditional FET + diode configuration can become a more efficient option. In case monolithic bidirectional switches are used, this crossover frequency increases to 470 kHz for the baseline operating conditions listed in Table III compared to 220 kHz when using hybrid bidirectional switches implemented with discrete power devices. Such an increase in the switching frequency up to 470 kHz with reduced losses compared to traditional CSIs is highly desirable for improving the power electronics power density by shrinking passive components such as the dc-link inductor as well as the cooling system.

VII. CONCLUSION

This article has presented and tested four-step and three-step commutation methods that enable the use of FQ WBG switches in CSIs, allowing substantial reduction of conduction losses in the power devices at the cost of a small increase in the switching losses. Experimental results confirm that the use of FQ switches can significantly boost the efficiency of CSIs topology. The three-step commutation method minimizes the switching loss overhead of using FQ switches by reducing the required time interval when the FQ switches operate in their lossy switch-diode mode by 3:1. The method does not add any mass, volume, or cost penalties.

Efficiency maps of several CSI topologies have been presented, highlighting the advantage of the three-step CSI implemented with monolithic bidirectional switches that can achieve an efficiency higher than 98.5% for most of the torque-speed range, while, this efficiency is mostly below 97.5% for the baseline FET + diode topology.

The datasheets of the newest generation of SiC power devices indicate that their losses are significantly reduced compared to the devices used for this investigation. As a result of this rapid progress, there is a reason for confidence that the CSI efficiency values reported in this article for 3 kW operation will be raised by 0.5 to 1% when these new devices are introduced. Ongoing research is aggressively pursuing these improvements, which will be reported in future papers.

ACKNOWLEDGMENT

The views and opinions of authors expressed herein do not necessarily state or reflect those of the United States Government or any agency thereof.

REFERENCES

- [1] E. A. Jones, F. F. Wang, and D. Costinett, "Review of commercial GaN power devices and GaN-based converter design challenges," *IEEE J. Emerg. Sel. Topics Power Electron.*, vol. 4, no. 3, pp. 707–719, Sep. 2016.
- [2] T. M. Jahns and H. Dai, "The past, present, and future of power electronics integration technology in motor drives," *CPSS Trans. Power Electron. Appl.*, vol. 2, no. 3, pp. 197–216, Sep. 2017.

- [3] H. Dai and T. M. Jahns, "Comparative investigation of PWM current-source inverters for future machine drives using high-frequency wide-bandgap power switches," in *Proc. IEEE Appl. Power Electron. Conf. Expo.*, 2018, pp. 2601–2608.
- [4] V. Madonna, G. Migliazza, P. Giangrande, E. Lorenzani, G. Buticchi, and M. Galea, "The rebirth of the current source inverter: Advantages for aerospace motor design," *IEEE Ind. Electron. Mag.*, vol. 13, no. 4, pp. 65–76, Dec. 2019.
- [5] V. Madonna, P. Giangrande, W. Zhao, H. Zhang, C. Gerada, and M. Galea, "On the design of partial discharge-free low voltage electrical machines," in *Proc. IEEE Int. Electric Mach. Drives Conf.*, 2019, pp. 1837–1842.
- [6] R. Cuzner, D. Drews, and G. Venkataramanan, "Power density and efficiency comparisons of system-compatible drive topologies," *IEEE Trans. Ind. Appl.*, vol. 51, no. 1, pp. 459–469, Jan. 2015.
- [7] S.-H. Ryu, "Monolithic bidirectional silicon carbide switching devices and methods of forming the same," US 2014/0027781 A1, Jan. 30, 2014.
- [8] S. Susumeci, M. Panizza, F. Stella, and F. Perraud, "Monolithic bidirectional switch based on GaN gate injection transistors," in *Proc. IEEE Int. Symp. Ind. Electron.*, 2020, pp. 1045–1050.
- [9] J. Honea, P. Parikh, Y. Wu, and I. Ben-Yaacov, "III-Nitride bidirectional switches," US 7,875,907 B2, Jan. 25, 2011.
- [10] M. Guacci *et al.*, "Three-phase two-third-PWM buck-boost current source inverter system employing dual-gate monolithic bidirectional GaN e-FETs," *CPSS Trans. Power Electron. Appl.*, vol. 4, no. 4, pp. 674–683, Dec. 2019.
- [11] N. Burany, "Safe control of four-quadrant switches," in *Proc. Conf. Rec. IEEE Ind. Appl. Soc. Annu. Meeting*, 1989, pp. 1190–1194.
- [12] H. Umeda *et al.*, "High power 3-phase to 3-phase matrix converter using dual-gate GaN bidirectional switches," in *Proc. IEEE Appl. Power Electron. Conf. Expo.*, San Antonio, TX, USA, 2018, pp. 894–897.
- [13] D. Han and B. Sarlioglu, "Deadtime effect on GaN-based synchronous boost converter and analytical model for optimal deadtime selection," *IEEE Trans. Power Electron.*, vol. 31, no. 1, pp. 601–612, Jan. 2016.
- [14] P. M. Roschatt, S. Pickering, and R. A. McMahon, "Bootstrap voltage and dead time behavior in GaN DC-DC buck converter with a negative gate voltage," *IEEE Trans. Power Electron.*, vol. 31, no. 10, pp. 7161–7170, Oct. 2016.
- [15] S. Ali and P. Wolfs, "An improved four step commutation process for silicon carbide based matrix converters," in *Proc. Australas. Universities Power Eng. Conf.*, 2016, pp. 1–5.
- [16] M. T. Fard, J. He, and Z. Wang, "Fault diagnosis and fault-tolerant operation of current source inverter for safety-critical applications," in *Proc. IEEE Transp. Electrification Conf. Expo.*, 2020, pp. 925–929.
- [17] Y. Bak, J. S. Lee, and K. B. Lee, "Balanced current control strategy for current source rectifier stage of indirect matrix converter under unbalanced grid voltage conditions," *Energies*, vol. 10, no. 1, 2017, Art. no. 27.
- [18] R. A. Torres, H. Dai, T. M. Jahns, and B. Sarlioglu, "Operation and analysis of current-source inverters using dual-gate four-quadrant wide-bandgap power switches," in *Proc. IEEE Energy Convers. Congr. Expo.*, 2019, pp. 2353–2360.
- [19] W. Lee, D. Han, W. Choi, and B. Sarlioglu, "Reducing reverse conduction and switching losses in GaN HEMT-based high-speed permanent magnet brushless dc motor drive," in *Proc. IEEE Energy Convers. Congr. Expo.*, 2017, pp. 3522–3528.
- [20] G. Ledwich, "Current source inverter modulation," *IEEE Trans. Power Electron.*, vol. 6, no. 4, pp. 618–623, Oct. 1991.
- [21] B. Wu, *High-Power Converters and AC Drives*. New York, NY, USA: IEEE Press, 2006.
- [22] H. Dai, R. A. Torres, W. Lee, T. M. Jahns, and B. Sarlioglu, "Integrated motor drive using soft-switching current-source inverters with SiC- and GaN-based bidirectional switches," in *Proc. IEEE Energy Convers. Congr. Expo.*, 2020, pp. 2372–2378.
- [23] Yokogawa Electric Corporation, Tokyo, Japan, *WT1800 Precision Power Analyzer Getting Started Guide*. (2011). [Online]. Available: <https://cdn.tmi.yokogawa.com/IMWT1801-03EN.pdf>
- [24] muRata, "muRata sim surfing," 2020. [Online]. Available: <https://www.murata.com/en-us/tool/simsurfing>
- [25] D. Neumayr, D. Bortis, J. W. Kolar, M. Koini, and J. Konrad, "Comprehensive large-signal performance analysis of ceramic capacitors for power pulsation buffers," in *Proc. IEEE Workshop Control Model. Power Electron.*, 2016, pp. 1–8.
- [26] C. B. Barth, T. Foulkes, I. Moon, Y. Lei, S. Qin, and R. C. N. Pilawa-Podgurski, "Experimental evaluation of capacitors for power buffering in single-phase power converters," *IEEE Trans. Power Electron.*, vol. 34, no. 8, pp. 7887–7899, Aug. 2019.
- [27] R. A. Torres, H. Dai, W. Lee, T. M. Jahns, and B. Sarlioglu, "A simple and robust controller design for high-frequency WBG-based current-source inverter-fed AC motor drive," in *Proc. IEEE Transp. Electrification Conf. Expo.*, 2020, pp. 111–117.
- [28] H. J. Lee, S. Jung, and S. K. Sul, "A current controller design for current source inverter-fed AC machine drive system," *IEEE Trans. Power Electron.*, vol. 28, no. 3, pp. 1366–1381, Mar. 2013.
- [29] R. A. Torres, H. Dai, T. M. Jahns, and B. Sarlioglu, "Design of high-performance toroidal DC-link inductor for current-source inverters," in *Proc. IEEE Appl. Power Electron. Conf. Expo.*, 2019, pp. 2694–2701.
- [30] Magnetics, "Edge material curves," 2021. [Online]. Available: <https://www.mag-inc.com/Products/Powder-Cores/Edge-Cores/Edge-Material-Curves>



Renato Amorim Torres (Member, IEEE) received the B.Sc. degree in electrical engineering from the Federal University of Minas Gerais, Belo Horizonte, Brazil, in 2016, and the M.S. degree in electrical engineering, in 2020 from the University of Wisconsin-Madison, Madison, WI, USA, where he is currently working toward the Ph.D. degree in electrical engineering.

Since 2017, he has been a Research Assistant with the Wisconsin Electric Machines and Power Electronics Consortium (WEMPEC), University of Wisconsin-Madison. His research interests include electric machines, power electronics with focus on machine controls, and wide-bandgap-based current-source inverters for integrated motor drive applications.



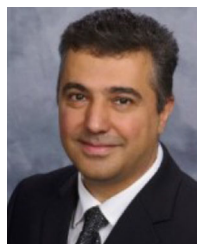
Hang Dai (Member, IEEE) received the B.Eng. degree in electrical engineering from North China Electric Power University, Beijing, China, in 2015, and the Ph.D. degree from the University of Wisconsin-Madison, Madison, WI, USA, in 2021, both in electrical engineering.

His research interests include power electronics and energy conversion systems.



Woongkul Lee (Member, IEEE) received the B.S. degree in electrical engineering from Yonsei University, Seoul, South Korea, in 2013, and the M.S. and Ph.D. degrees in electrical engineering from the University of Wisconsin-Madison, Madison, WI, USA, in 2016 and 2019, respectively.

From 2019 to 2020, he was a Postdoctoral Research Associate with the Wisconsin Electric Machines and Power Electronics Consortium (WEMPEC), University of Wisconsin-Madison. In 2020, he joined the Department of Electrical and Computer Engineering, Michigan State University, East Lansing, MI, USA, as an Assistant Professor. His research interests include wide bandgap device-based power electronics, high-performance motor drive, and multilevel or multiphase motor drives.



Bulent Sarlioglu (Senior Member, IEEE) received the B.S. degree in electrical engineering from Istanbul Technical University, Istanbul, Turkey, the master's degree in electrical engineering from the University of Missouri, Columbia, MO, USA, and the Ph.D. degree in electrical engineering from the University of Wisconsin-Madison, Madison, WI, USA.

Since 2011, he has been a Professor with the University of Wisconsin-Madison and the Associate Director of the Wisconsin Electric Machines and Power Electronics Consortium. From 2000 to 2011, he was with the Aerospace Division, Honeywell International Inc., Torrance, CA, USA, as a Staff Systems Engineer. He has authored or coauthored more than 200 technical papers that are published in conference proceedings and journals, and is the Inventor or Co-Inventor of 20 U.S. patents and many international patents. His research interests include electrical machines, drives, and power electronics.

Dr. Sarlioglu was the recipient of the Honeywell's Outstanding Engineer Award in 2011, the NSF CAREER Award in 2016, and the 4th Grand Nagamori Award from Nagamori Foundation, Japan, in 2018.



Thomas M. Jahns (Fellow, IEEE) received the S.B., S.M., and Ph.D. degrees in electrical engineering from the Massachusetts Institute of Technology, Cambridge, MA, USA, in 1974, 1974, and 1978, respectively.

In 1998, he joined as a Grainger Professor of Power Electronics and Electric Machines with the Department of Electrical and Computer Engineering, University of Wisconsin-Madison, Madison, WI, USA, where he is currently a Co-Director with the Wisconsin Electric Machines and Power Electronics Consortium (WEMPEC). Prior to joining the University of Wisconsin-Madison, he was with GE Corporate Research and Development (now GE Global Research Center), Niskayuna, NY, USA, for 15 years. From 1996 to 1998, he conducted a research sabbatical with MIT, Cambridge, MA, USA, where he directed research in advanced automotive electrical systems as a Co-Director with an industry-sponsored consortium. His current research interests include high-performance permanent-magnet synchronous machines, integrated motor drives, and applications of wide-bandgap power switches to machine drives.

Dr. Jahns was elected as a Member of the US National Academy of Engineering in 2015. He was a Distinguished Lecturer of the IEEE Industry Applications Society and Power Electronics Society (PELS). He is a Past President of PELS, and from 2001 to 2002, he was the Division II Director on the IEEE Board of Directors. He was the recipient of the 2005 IEEE Nikola Tesla Technical Field Award and the IAS Outstanding Achievement Award in 2011.